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FEDERAL SPECIFICATION

STEEL BARS, CARBON, COLD FINISHED AND HOT ROLLED, (GENERAL PURPOSE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers carbon steel bars in the forms used for general fabrication purposes (see 6.1).

1.2 Classification.

1.2.1 *Composition.*—Steel bars shall be furnished in the composition listed in table I, as specified.

1.2.2 *Condition.*—Steel bars shall be furnished hot rolled or cold finished in the conditions as specified (see 6.2). If no condition is specified, hot rolled steel bars shall be furnished as rolled or annealed and cold finished steel bars shall be furnished in the as cold finished condition (either as cold rolled, as cold drawn, or turned and polished).

Hot rolled bars:

HR—As hot rolled.
ANL—Annealed.
NORM—Normalized.
NT—Normalized and tempered.
QT—Quenched and tempered.

Cold finished bars:

CF—As cold finished (as cold drawn or as cold rolled).
CR—As cold rolled.
CD—As cold drawn.
ACF—Annealed and cold finished.
NCF—Normalized and cold finished.
NCFSR—Normalized, cold finished and stress relieved.
CFSR—Cold finished and stress relieved.
TP—Turned and polished.
CDGP—Cold drawn, ground and polished.
TGP—Turned, ground and polished.

1.2.3 *Form.*—Steel bars shall be furnished in the following forms, as specified (see 6.2):

Round.
Square.
Flat.
Hexagonal.

1.2.3.1 In addition, hot rolled bars shall also include oval and half-oval, half-round bars ½ inch or more in thickness or diameter, and bar-sized shapes.

1.3 Quality.—Hot rolled steel bars shall be furnished as special bar quality unless merchant bar quality is specified. Cold finished bars shall be furnished as special bar quality (see 3.1 and 6.2).

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2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 The following standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Standards:

Fed. Std. No. 48—Tolerances for Steel and Iron Wrought Products.

Fed. Std. No. 66—Steel: Chemical Composition and Hardenability.

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL–STD–163 — Preparation of Steel Products for Domestic Shipment (Storage and Overseas Shipment).

MIL–STD–183 — Continuous Identification Marking of Iron and Steel Products.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** — The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Consolidated Classification Committee Publication:

Consolidated Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Manufacture.—The steel shall be produced by either the open hearth or the electric furnace process, unless one of these two processes is specified, except steels in the B11XX series shall be made by the Bessemer Process. Unless otherwise specified and except for steels in the B11XX and C12XX series (see 6.2.2) the steel shall be killed or semi-killed. Merchant quality bar may be produced from rimmed, capped, or semi-killed steel at the option of the producer.

3.2 Chemical composition.—The steel numbers designating the compositions covered by this specification are listed in table I. The chemical composition for the specified steel number shall meet the requirements of Federal Standard No. 66 (see 3.2.1 and 3.2.2).

3.2.1 *Ladle analysis.*—A ladle analysis of each heat or lot of steel shall be furnished by the contractor, showing the percentage of the elements present. The ladle analysis shall meet the requirements of the specified steel number in Federal Standard No. 66.

3.2.2 *Check analysis.*—The chemical composition as determined by analysis shall meet the requirements specified in Federal Standard No. 66.

3.2.3 *B11XX and C12XX series steels.*—Steels in the B11XX or C12XX series with identical chemical composition shall be considered equivalent for bars covered by this

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TABLE I.—*Steel composition numbers for applicable carbon steels*[a]

Steel number	Steel number	Steel number	Steel number	Steel number	Steel number
Standard nonresulphurized steels					
C1008	C1019	C1027	C1038	C1049	C1078
C1010	C1020	C1029	C1039	C1050	C1080
C1011	C1021	C1030	C1040	C1051	C1084
C1012	C1022	C1031	C1041	C1052	C1085
C1015	C1023	C1033	C1042	C1053	C1086
C1016	C1024	C1035	C1043	C1055	C1090
C1017	C1025	C1036	C1045	C1060	C1095
C1018	C1026	C1037	C1046	C1070	
Basic open hearth resulphurized[b] steels					
C1108	C1115	C1119	C1132	C1140	C1145
C1109	C1116	C1120	C1137	C1141	C1146
C1110	C1117	C1125	C1138	C1144	C1151
C1113	C1118	C1126	C1139		

Basic open hearth rephosphorized and resulphurized steels	Acid bessemer resulphurized steels
C1211	B1111
C1212	B1112
C1213	B1113

[a] Fed. Std. No. 66 designates the compositions that are used more frequently and are in general more readily available.

[b] Also known as free cutting steels.

specification, subject to agreement between purchaser and contractor. Unless otherwise specified, when a steel in either the B11XX series or C12XX series is specified, the equivalent steel in either series may be furnished, provided the contractor has given the purchasing agency prior notice and has secured agreement on the interchangeability.

3.3 Mechanical properties.—Steel bars shall meet the mechanical property requirements as specified by the procuring agency when tested in accordance with 4.4.2.

3.3.1 *Heat treatment.*—Steel bars, specified in one of the heat-treated conditions (see 1.2.2) shall meet the mechanical property requirements as specified by the procuring agency when tested in accordance with 4.4.2. The contractor may determine the detail procedure to produce the required mechanical properties throughout the entire lot. Steel bars which are straightened following heat-treatment may be stress relieved, if necessary, to meet the specified properties.

3.4 Hardness.—When specified, steel bars shall be furnished to hardness values as specified by the procuring agency when tested in accordance with 4.4.3.

3.5 Dimensions and tolerances.—For all dimensional tolerances the requirements of Federal Standard No. 48 shall apply. The applicable references to this standard are listed in table II (see also 6.2.2).

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TABLE II.—*Dimensional tolerance reference to Federal Standard No. 48*

Form	Dimensional tolerance	Fed. Std. No. 48 Paragraph reference	
		Hot rolled	Cold finished
Rounds	Size and diameter, out-of-round	1a1	2a1, 2a2
Squares	Size and width, out-of-square	1a1	2a1
Hexagons	Size, out-of-hexagon	1a2	2a1
Flats	Thickness and width	1a3	2a1
Bar size angles	Thickness, length and out-of-square	1a4	---
Bar size channels	Thickness of web and out-of-square	1a5	---
Bar size tees	Width, thickness and stem out-of-square	1a6	---
Other special bar size sections	All	1a7	---
All	Length	1a11	---
All	Random lengths	1a8	2a3
All	Specific lengths	1a9, 1a10	2a4
All	Straightness[1]	1a12	2a5

[1] Hot rolled bars ordered special straightened shall be straight within ⅛-inch in any 5 feet of length, or within ⅛-inch times the number of feet divided by 5, in the entire length.

3.6 Macro-examination. — When specified, steel bars shall meet standards of acceptance or rejection for macro-examination as specified by the procuring agency when tested in conformance with 4.4.4 (see 6.2).

3.7 Magnetic particle inspection. — When specified, cold finished steel bars shall meet the standards of acceptance or rejection as specified by the procuring agency when tested in accordance with 4.4.5 (see 6.2).

3.8 Finish.

3.8.1 ***Hot rolled and hot rolled, heat treated bars.***—Steel bars shall be commercially free from flat spots, deep rolling marks and excessive scale.

3.8.1.1 ***Scale free bars.*** — When specified, steel bars shall be cleaned of scale by pickling or other approved method.

3.8.2 ***Cold finished and heat treated cold finished bars.***—Steel bars, cold finished, shall have a smooth unoxidized surface. Steel bars, cold finished and heat treated or stress relieved shall be free from scale, but may have the characteristic oxidized surface.

3.9 Identification marking. — When specified, steel bars shall be marked for identification in accordance with Military Standard MIL–STD–183. For Navy, Bureau of Aeronautics, all steel bars shall be marked in accordance with Military Standard MIL–STD–183.

3.10 Workmanship.

3.10.1 Steel bars shall be free from pipes, flakes, and heat checks, and shall not contain any other defects which, due to their nature, degree or extent, shall affect detrimentally the suitability of the bars for their intended use.

3.10.2 Each length of steel bars shall consist of one continuous piece as cut from the full length bar as rolled. Bars joined together by welding shall not be furnished.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.—Unless otherwise specified, a lot shall consist of all bars submitted for inspection at the same time, of the same heat, the same condition, the same finish, the same diameter or thickness, and when heat-treated from the same process and the same batch.

4.2 Sampling.

4.2.1 *Samples for Government chemical check analysis.*—At least one sample for Government analysis shall be taken from bars from each heat. Samples shall be composed of chips taken from bars selected by the inspector and shall be free from oil, grease, dirt or other foreign material. The sample shall weigh at least 5 ounces. The samples obtained shall be placed in separate containers, and forwarded to a Government or Government approved laboratory as designated by the procuring agency.

4.2.2 *Samples for mechanical tests.*—When mechanical properties are specified, at least one test sample shall be selected for each 10,000 pounds of material or fraction thereof in each lot, provided that not fewer than two samples shall be taken for any one lot and not more than ten samples shall be taken for any lot. In addition, in the case of bars specified in the quenched and tempered condition, if more than one quenching charge is represented in a tempering charge, each quenching charge shall be represented in the material selected for testing.

4.2.3 *Samples for hardness tests.*—When hardness is specified, at least one sample shall be selected for each 10,000 pounds of material, or fraction thereof, in each lot, provided not fewer than five tests shall be made for each lot.

4.2.4 *Samples for macro-examination tests.*—When specified, macro-examination shall be made on representative billets from which the steel will be rolled. In the event small quantities of bars are secured from warehouse stock, the macro-examination shall be made on samples cut from the bars, provided the identity of the material can be established to the satisfaction of the inspector.

4.2.4.1 When macro-examination is made of billets, a full cross section sample, approximately ½ inch thick, shall be taken from the top end of the top billet and from the bottom end of the bottom billet of the first, middle and last useable ingot of the heat.

4.2.4.2 When macro-examination is made of bars, a full cross section sample, approximately ½ inch thick, shall be taken from each end of each bar selected for the tension test, or in cases where a tension test is not required, from an equivalent number of bars.

4.2.5 *Samples for visual and dimensional examination.*—When inspection by attributes is specified, a random sample shall be selected by the inspector from each lot for visual and dimensional examination. The number of pieces in the random sample shall be in accordance with table III.

TABLE III.—*Sampling for visual and dimensional examination AQL (approx.) = 1.5 percent defective*

Number of pieces in lot	Number of pieces in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

4.3 Examination.

4.3.1 *Visual and dimensional examination.*

4.3.1.1 Unless inspection by attributes is specified, all steel bars shall be examined

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for compliance with the workmanship, finish, identification marking, and dimensional requirements of this specification.

4.3.1.2 When inspection by attributes is specified, each of the sample pieces selected in accordance with 4.2.5 shall be examined visually and dimensionally to determine compliance with the workmanship, finish, identification marking and dimensional requirements of this specification.

4.3.1.3 Any nonconforming piece in each sample shall be rejected and if the number of nonconforming pieces in each sample is equal to or greater than the rejection number shown in table III for the size of lot under examination, the lot represented by the sample shall be rejected.

4.3.1.4 Lots rejected on the basis of visual or dimensional defects may be resubmitted for inspection, providing the contractor has removed all nonconforming pieces.

4.3.2 *Preservation, packaging, packing, and shipment marking examination.*—The inspector shall ascertain that the steel bars are preserved, packaged, packed and marked for shipment in conformance with section 5.

4.4 Test procedures.

4.4.1 *Chemical analysis.*—The samples selected and forwarded in accordance with 4.2.1 shall be analyzed in conformance with methods 111 or 112 of Federal Test Method Standard No. 151. In case of dispute, method 111 shall be used for referee analysis.

4.4.2 *Tension tests.*—Specimens selected in accordance with 4.2.2 shall be tested in accordance with method 211 of Federal Test Method Standard No. 151 as follows.

4.4.2.1 Bars under 0.75 inch in diameter or thickness may be tested in full section when practicable. Larger bars may be tested in full size provided an extensometer is not required in determining elastic properties. In the latter case, machined test specimens shall be used. Machined test specimens shall conform to the form and dimensions of the appropriate test specimen described in method 211 of Federal Test Method Standard No. 151, using the largest specimen type practicable.

4.4.2.2 The location of test specimens shall be as shown in table IV.

TABLE IV.—*Location of tension test specimens*

Form	Size (Diameter or thickness of bar; width of flat)	Location of axis of test specimen
Bars, flats	Not over 1½ inches	Coincides with central axis of piece
Bars	Over 1½ inches	One-fourth the diameter or thickness from the surface, and parallel to the direction of rolling
Flats	Over 1½ inches	Test bars located at one-fourth width, and parallel to direction of rolling

4.4.3 *Hardness tests.*—Hardness tests on samples selected in accordance with 4.2.3 shall be made in accordance with method 242 or 243 of Federal Test Method Standard No. 151 as specified by the procuring agency.

4.4.4 *Macro-examination tests.* — Macro-examination tests on samples selected in accordance with 4.2.4 shall be conducted in accordance with method 321 of Federal Test Method Standard No. 151.

4.4.5 *Magnetic particle inspection.*—When specified, magnetic particle inspection shall be made on all bars in accordance with the method specified by the procuring agency.

4.5 Rejection and retest.

4.5.1 *Rejection.*—If a test specimen fails to meet any of the requirements of this specification, the lot represented by the specimen shall be rejected.

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4.5.2 *Retests.*—Retests shall be permitted in accordance with Federal Test Method Standard No. 151.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.*—Preservation and packaging of steel bars shall be in accordance with Military Standard MIL–STD–163.

5.1.2 *Level C.*—Preservation and packaging shall be in accordance with standard commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Packing of steel bars shall be in accordance with Military Standard MIL–STD–163.

5.2.2 *Level C.*—Packing shall be in accordance with standard commercial practice adequate to insure carrier acceptance and safe delivery at the lowest rate. Shipments shall comply with the requirements of the Consolidated Freight Classifications Rules or other regulations applicable to the mode of transportation (see 6.2).

5.3 Marking for shipment.—Marking for shipment shall be in accordance with Military Standard MIL–STD–163.

6. NOTES

6.1 Intended use. — Steel bars procured under this specification are intended for general fabricating purposes, excluding concrete reinforcing bars.

6.2 Ordering data. — Procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Composition, condition and form (see 1.2).
(*c*) Quality required (see 1.3).
(*d*) Size (see 3.5).
(*e*) Selection of hardness values (see 3.4).
(*f*) Selection of mechanical properties (see 3.3).
(*g*) Requirements for macro-examination (see 3.6).
(*h*) Requirements for magnetic inspection (see 3.7).
(*i*) Finish required (see 1.2.2 and 3.8).
(*j*) Whether inspection by attributes is required (see 4.2.5).
(*k*) Level of preservation, packaging and packing required (see section 5).

Federal Standard No. 102 should be referred to for definitions of various levels of preservation, packaging and packing protection.

6.2.1 *Silicon content.*—Open hearth steels may be ordered to a maximum limit or range of silicon as shown in Federal Standard No. 66. In such cases the specifications should clearly indicate the selected limit or range and state that bars shall meet the limits indicated. Acid bessemer steel and basic open hearth rephosphorized carbon steels are not furnished with specified silicon contents.

6.2.2 *Multiple lengths.*—When bars are ordered in lengths expressed as multiples of specific units, the amount required for any loss in parting the units should be included in the lengths delivered.

6.3 Definition.

6.3.1 *Bar.* — The definition of the term "bar", on the basis of the classification by size of flat rolled, hot rolled steels is 0.2299 inches or over in thickness and up to 6 inches in width.

6.3.2 *Definition of bar-sized shapes.*—Bar-sized shapes are angles, channels, tees, and zees with greatest cross-sectional dimensions less than 3 inches.

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6.4 Definitions and other information.—Department of Defense, Supply and Logistics Handbook, Standardization H–8, Steel and Wrought Iron Products may be consulted for guidance, general information, and definitions. The handbook is not intended for use for procurement and is limited to informational purposes only.

6.5 Transportation description.—The transportation description applicable to the bars covered by this specification is as follows:

Bars, steel, not otherwise indexed, loose or in packages:

Carload minimum weight, 40,000 pounds.

Truckload minimum weight, 36,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—O C E Q Sig T
Navy—Sh A Or Y MC
Air Force.

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